

# EMIR BEKAR

Mechatronics Engineer · Game Developer · Content Creator

Istanbul, Turkey   ✉ [emir.bekar@bilgi.edu.net](mailto:emir.bekar@bilgi.edu.net)   [in linkedin.com/in/emir-bekar-883410360](https://www.linkedin.com/in/emir-bekar-883410360)   [github.com/emiukob](https://github.com/emiukob)

★ LIVE PLAYABLE MAP  
[emiukob.github.io](https://emiukob.github.io) ↗

## SUMMARY

Mechatronics Engineering graduate (100% English) with strong mathematical and analytical foundations, award-winning senior design research, and two IEEE-published papers in multi-agent robotics and AI. Hands-on experience developing mobile 3D/2D games in Unity, creating custom 2D pixel art and 3D assets, and managing an active 12,000+ subscriber YouTube community with direct insight into player retention and content iteration. Passionate about mobile casual games and eager to drive engaging product features and analytics at Peak.

## EDUCATION

### Istanbul Bilgi University

Jul 2026

Bachelor of Science in Mechatronics Engineering (100% English)

Istanbul, Turkey

- 🏆 **Best Senior Design Project Award:** "Adaptive Formation Control for Collaborative Swarm Robots" (Presented at HORA Congress 2026, published on IEEE Xplore).

## FEATURED PROJECTS

### Match 3D Mobile Game

Unity 6 · C# · Blender · URP

- Developed a hybrid-casual 3D matching mobile game with physics-based matching and zero-GC object pooling for stable 60 FPS.
- Designed core retention and monetization meta-game loops: Daily Rewards, Lucky Spin wheel, and in-game economy store.
- Modeled and textured modular 3D assets in Blender for mobile visual clarity. [[GitHub](#) | [Video Demo](#) ↗]

### 2D Pixel Art Indie Game (In Development)

Unity 2D · C# · LibreSprite · Pixel Art

- Hand-crafting original 2D pixel art character sprites, frame-by-frame animations, environment tilesets, and modular visual effects.
- Programming responsive 2D physics controllers, state machines, and interactive world gameplay mechanics from scratch.

### AC-130 Gunship Simulation (In Development)

Unity 6 · C# · Custom Shaders

- Close air support tactical simulation featuring custom FLIR thermal vision shaders, dynamic terrain destruction, and avionics HUD telemetry. [[GitHub](#) | [Video Demo](#) ↗]

## PUBLICATIONS (IEEE XPLORE · HORA CONGRESS 2026)

- Adaptive Formation Control for Collaborative Swarm Robots (IEEE 2026):** Leader-follower formation switching via FSM and fuzzy-logic ACC control eliminating stop-and-go oscillations, validated in Webots simulation. [[IEEE Xplore #11537161](#) ↗]
- Multi Agent Reinforcement Learning Based Swarm Navigation (IEEE 2026):** Autonomous swarm coordination and collision-free path optimization using Multi-Agent Reinforcement Learning (MARL). [[IEEE Xplore #11537199](#) ↗]

## COMMUNITY & CONTENT CREATION

### Emiukob – YouTube Creator & Mod Developer

12,000+ Subscribers · [youtube.com/@emiukob](https://youtube.com/@emiukob)

- Built a gaming and tech community to 12,000+ subscribers; analyzed audience retention curves, drop-off rates, and CTR to iteratively optimize content pacing.
- Developed and published "Hologram Projector" 3D hologram Minecraft mod in Java on CurseForge. [[CurseForge](#) ↗]

## SKILLS & CERTIFICATIONS

**Game & Product:** Mobile casual / hybrid-casual games, 3D match mechanics, level pacing, FTUE, player retention & meta-game loops

**Art & Technical Tools:** Unity 6 / 2D (C#), 2D Pixel Art & Animation, LibreSprite, Blender (3D), Python, Java, JavaScript, Git/GitHub

**Analytical:** Data interpretation, A/B testing mindset, mathematical modeling, fuzzy logic, reinforcement learning, Webots

**Certifications:** Basic Python for Business Analytics (Arizona State University) | UAV 0/1 Pilot License (SHGM)